AI/ML intern DAY 6

TASK 4 — CREATING A BASIC FASHION RECOMMENDATION DATASET

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1. Abstract

This task focused on creating a basic AI fashion recommendation dataset and understanding how recommendation systems use customer preferences and fashion patterns to generate intelligent outfit suggestions.

The task explored:

- customer preference analysis
- fashion dataset creation
- outfit recommendation logic
- AI shopping workflows
- intelligent clothing prediction

The research also studied how AI systems analyse fashion trends and customer behaviour to recommend suitable clothing combinations.

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2. Objective

The objective of this task was to create a simple fashion recommendation dataset and understand how AI systems generate personalized clothing suggestions.

The task focused on:

- fashion dataset creation
- recommendation logic
- customer preference analysis

- outfit categorization
- intelligent shopping systems
- fashion prediction workflows

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3. Introduction to AI Fashion Recommendation Systems

AI fashion recommendation systems are intelligent applications that suggest clothing items based on:

- customer preferences
- shopping history
- clothing patterns
- colour choices
- fashion trends
- body type

The AI system analyses customer behaviour and predicts suitable outfits dynamically.

These systems are commonly used in:

- online shopping platforms
- fashion recommendation systems
- AI shopping assistants
- digital wardrobe systems
- smart retail applications

t-SNE representation of some dresses and shoes projected into style space. The blue box highlights an area containing occasion wear while the red box contains casual day wear.

Figure 1 — AI Fashion Recommendation System

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4. Structure of the Recommendation Dataset

The basic fashion recommendation dataset contains multiple customer preference fields.

Dataset Field	Purpose
Customer ID	User identification
Preferred Style	Fashion category
Favourite Colour	Colour preference
Clothing Type	Outfit selection
Season	Weather analysis

5. Sample Fashion Recommendation Dataset

Customer ID	Preferred Style	Favourite Colour	Season	Recommended Outfit
101	Casual	Black	Winter	Hoodie + Jeans
102	Formal	Blue	Summer	Shirt + Trousers
103	Streetwear	White	Winter	Oversized Hoodie
104	Sportswear	Red	Summer	Tracksuit
105	Traditional	Green	Festival	Kurta

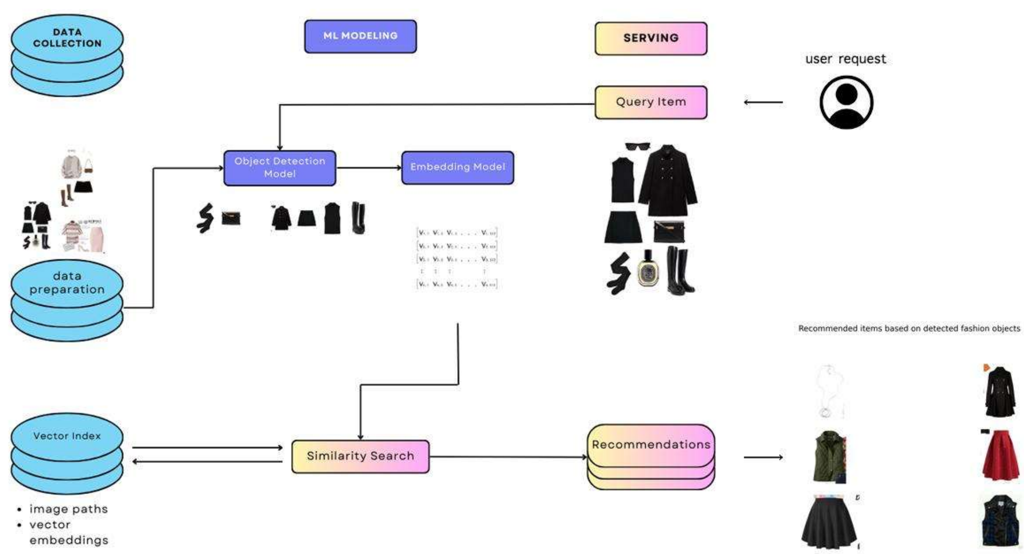

Figure 2 — Fashion Dataset Workflow

6. Workflow of AI Recommendation Systems

The recommendation workflow consists of multiple AI processing stages.

Stage 1 — Customer Data Collection

The AI system collects:

- fashion preferences
- color interests
- shopping history
- body type
- seasonal interests

Stage 2 — Behaviour Analysis

The AI analyses:

- customer searches
- frequently selected outfits
- purchase history
- outfit combinations

Stage 3 — Pattern Recognition

Machine learning models identify:

- fashion patterns
- color compatibility
- style categories
- customer interests

Stage 4 — Recommendation Generation

The AI system generates:

- matching outfits
- seasonal recommendations
- personalized clothing suggestions
- smart fashion combinations

Stage 5 — Final Output

The final recommended outfit is displayed to the customer dynamically.

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7. Basic Recommendation Logic

The AI system follows simple recommendation logic based on customer interests.

Casual Recommendation Logic

IF:

- customer prefers hoodies
- customer likes sneakers
- customer searches casual outfits

THEN:

- recommend streetwear outfits
- suggest cargo pants
- recommend matching sneakers

Formal Recommendation Logic

IF:

- customer prefers blazers
- customer selects office wear
- customer likes dark colours

THEN:

- recommend formal shirts
- suggest trousers
- recommend formal shoes

Seasonal Recommendation Logic

IF:

- season is winter

THEN:

- recommend hoodies
- jackets
- boots
- layered outfits

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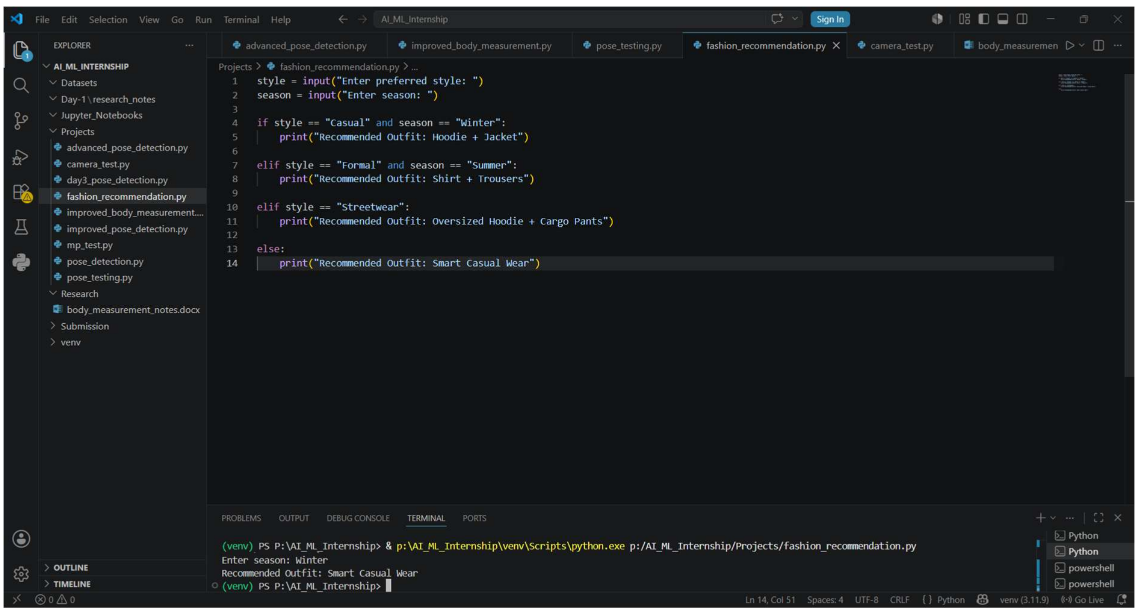
8. Simple Python Recommendation Example



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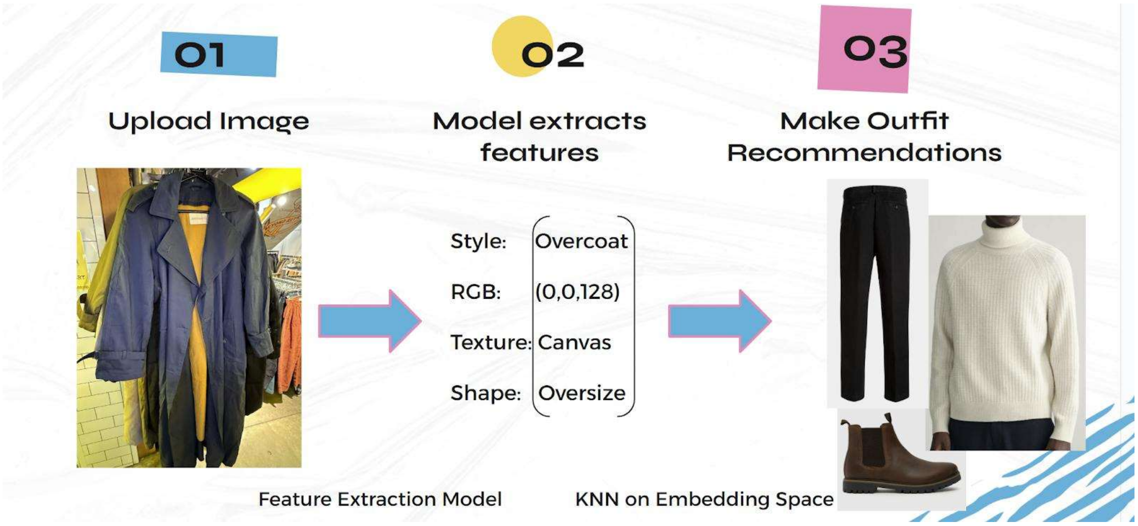
9. Technologies Used

Technology	Purpose
Artificial Intelligence	Recommendation generation
Machine Learning	Pattern analysis
Python	Dataset implementation
Computer Vision	Fashion analysis
Deep Learning	Recommendation improvement



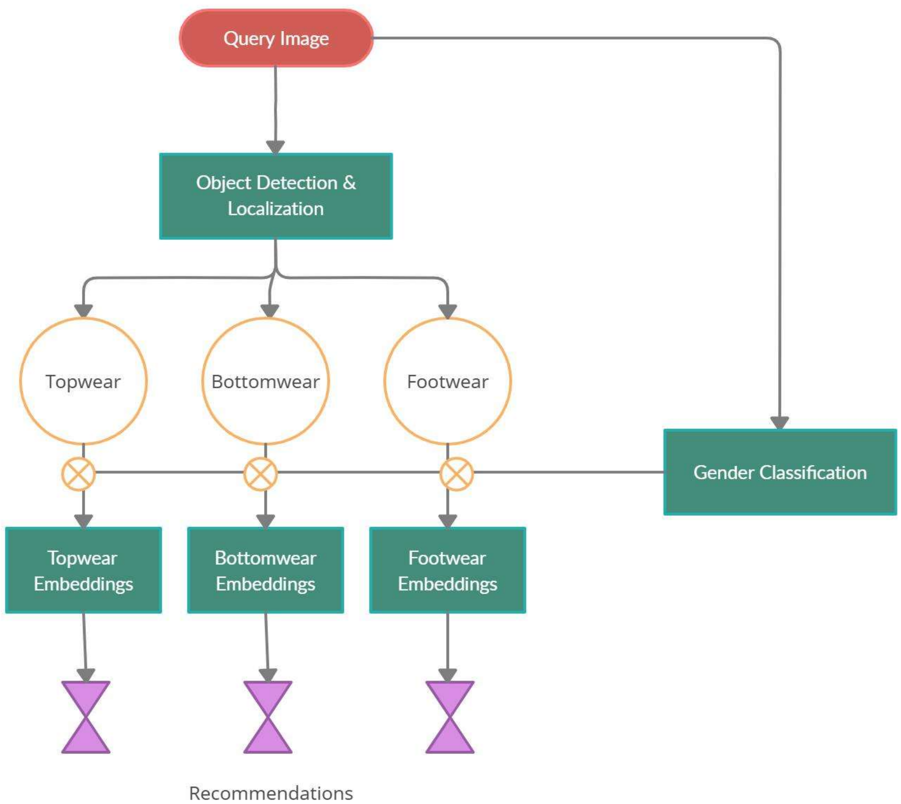


Figure 3 — AI Outfit Recommendation Workflow

10. Applications of AI Fashion Recommendation Systems

AI recommendation systems are used in:

- online shopping platforms
- AI shopping assistants
- digital wardrobe systems
- personalized fashion systems
- intelligent retail systems
- virtual shopping applications

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11. Challenges Faced

Challenge	Impact
Limited customer data	Reduces recommendation accuracy
Changing fashion trends	Requires continuous learning
Incorrect outfit categorization	Affects recommendation quality
Diverse customer preferences	Complex recommendation logic

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12. Knowledge and Skills Acquired

During this task, the following concepts were understood:

- recommendation dataset creation
- AI fashion workflows
- customer preference analysis
- intelligent shopping systems
- fashion prediction logic
- outfit recommendation systems
- machine learning recommendation workflows

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13. Conclusion

AI fashion recommendation systems improve online shopping experiences using Artificial Intelligence and Machine Learning technologies.

The task improved understanding of:

- recommendation dataset creation
- AI shopping systems
- outfit recommendation workflows
- fashion pattern analysis
- customer behaviour analysis
- intelligent recommendation systems

These systems are widely used in modern fashion technology, e-commerce platforms, and personalized shopping applications.